

Appendix

A Appendix.A Data Setting

The experiments were conducted on three well-known topic classification datasets and three sentiment classification datasets. Table ?? provides detailed statistics for each dataset. Detailed descriptions of each dataset are as follows:

AG’s News¹: The experiments were conducted on the AG’s News dataset for news topic classification. This dataset was constructed by selecting the four largest categories from the original AG’s News corpus. The AG’s News dataset comprises 120,000 English news articles from over 2,000 news sources, categorized into four major classes: World, Sports, Business, and Science/Technology. It is important to note that, in the experiments, only the news headlines from AG’s News were chosen as short texts.

Snippets²: The Snippets dataset consists of three components: url, title, and text description. It is composed of web search fragments collected by Google in response to query phrases submitted to the Google search engine. The class labels of the collected search fragments are assigned the same class labels as the query phrases.

TMN News³: Similar to the AG’s News dataset, the TMN News dataset also consists of English news articles from popular newspaper websites. It includes 32,000 English news articles across seven different categories. Notably, similar to the AG’s News dataset, only news headlines are used as input texts in the experiments.

Twitter⁴: Sample tweets were downloaded from the Natural Language Toolkit (NLTK), which includes 20,000 tweet samples retrieved from the Twitter stream API. This dataset is utilized for news sentimental classification, comprising 5,000 tweets with negative sentiment and 5,000 tweets with positive sentiment.

MR⁵: The MR (Movie Reviews) dataset is a corpus designed for sentiment analysis experiments. It includes movie review documents annotated based on overall sentiment polarity (positive or negative) or subjective ratings (e.g., ”2.5

stars”). Additionally, the dataset contains sentences labeled for subjectivity (subjective or objective) and polarity.

SST-2⁶: The SST-2 (Stanford Sentiment Treebank 2) sentiment analysis dataset is a widely used corpus containing movie review snippets annotated with positive or negative sentiment labels, utilized for training and evaluating text sentiment classification models.

B Appendix.B Baseline Methods

To validate the effectiveness of our approach, we conducted comparisons with SOTA methods.

BERT[Devlin *et al.*, 2019] BERT is a bidirectional encoder representation method based on the Transformer architecture. It transforms short text classification tasks into cloze-style fill-in-the-blank questions and serves as an effective baseline method.

Prompt[Liu *et al.*, 2023a] Regular Prompt-tuning involves filling input sentences into hand-crafted templates and forming label vocabulary space using only category names. For fairness, PL employs manual templates that align with our method.

P-tuning[Liu *et al.*, 2023b] The method proposes learning continuous prompts by inserting trainable variables into embedded inputs rather than using manual templates.

Yi-34B[AI *et al.*, 2024] Yi-34B is one of the next-generation open-source large language model series developed from scratch by 01.AI. It is designed for bilingual use and trained on a 3T multilingual corpus. Yi-34B excels in language understanding, commonsense reasoning, and reading comprehension, making it one of the most powerful large language models globally.

LLaMA3[Touvron *et al.*, 2023] An advanced large language model developed by Meta AI, employing an optimized Transformer architecture. It is an autoregressive language model designed for various natural language processing tasks.

GPT-4[OpenAI, 2023] A highly advanced language model developed by OpenAI, known for its extensive capabilities in natural language understanding and generation tasks, building upon the Transformer architecture.

ZeroGen[Ye *et al.*, 2022] An efficient zero-shot learning method that leverages pre-trained language models to unsupervisedly generate datasets. Subsequently, it trains small

¹http://www.di.unipi.it/~gulli/AG_corpus_of_news_articles.html

²<http://jwebpro.sourceforge.net/data-web-snippets.tar.gz>

³<http://acube.di.unipi.it/tmn-dataset/>

⁴https://raw.githubusercontent.com/nltk/nltk_data/gh-pages/packages/corpora/twitter_samples.zip

⁵https://huggingface.co/datasets/cornell-movie-review-data/rotten_tomatoes

⁶<https://nlp.stanford.edu/sentiment/index.html>

task-specific models on these synthetic datasets, achieving efficient inference and benefiting from not requiring annotated data.

GenCo[Zhang *et al.*, 2024] A method leveraging the generative capability of large-scale language models to aid in training smaller, more adaptable language models. This approach utilizes semantic augmentation and generates high-quality pseudo-labels, demonstrating superior performance on limited-domain data.

C Appendix.C Examples of Prompts Generated by LLMs

Tables 1 and 2 respectively present the various input prompts generated by LLMs for topic classification and sentiment classification tasks. Each task generated a total of 15 prompts, with [MASK] representing the masked character.

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[Zhang *et al.*, 2024] Ruohong Zhang, Yau-Shian Wang, and Yiming Yang. Generation-driven contrastive self-training for zero-shot text classification with instruction-following llm. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 659–673, 2024.

Table 1: Topic Classification Prompts Generated via GPT-4.

Input:	This is a news about [MASK].
1	Breaking news: A report just in about [MASK].
2	Bringing you the latest news regarding [MASK].
3	Stay up-to-date with this newflash on [MASK].
4	Get the scoop on the latest events involving [MASK].
5	Tune in for the latest updates on [MASK].
6	Fresh off the press: A news bulletin covering [MASK].
7	Discover the latest developments surrounding [MASK].
8	Keep your finger on the pulse with this report on [MASK].
9	Unveiling the details of the news story about [MASK].
10	Dive into this exclusive coverage of [MASK].
11	Learn about the unfolding events related to [MASK].
12	Catch up on the latest happenings with this news report on [MASK].
13	Stay in the know with this comprehensive coverage of [MASK].
14	Explore the intricacies of this breaking news story concerning [MASK].
15	Follow along as we uncover the latest updates on [MASK].

Table 2: Sentiment Classification Prompts Generated via GPT-4.

Input:	This is a comment that expresses [MASK] emotion.
1	The sentiment conveyed by this sentence is [MASK].
2	This sentence encapsulates the feeling of [MASK].
3	Delve into the emotional depth of this sentence: [MASK].
4	Take a moment to experience the heartfelt emotion expressed in this sentence: [MASK].
5	Explore the mood evoked by this sentence, which is [MASK].
6	This sentence radiates with the essence of [MASK] emotion.
7	Experience the raw emotion captured within this sentence: [MASK].
8	Behold the sentiment conveyed in this sentence, which is [MASK].
9	Connect with the underlying emotion expressed in this sentence: [MASK].
10	This sentence speaks volumes about the [MASK] feeling it conveys.
11	Immerse yourself in the depth of emotion articulated in this sentence: [MASK].
12	Engage with the sincerity of [MASK] emotion expressed in this sentence.
13	Witness the passion evident in this sentence, imbued with [MASK] emotion.
14	Feel the warmth of [MASK] emotion emanating from this sentence.
15	Dive into the emotional landscape painted by this sentence, revealing [MASK].